



# On the verge of a digital divide in the use of generative AI?☆

David Suárez<sup>\*</sup>, Begoña García-Mariño

Department of Statistics and Knowledge Management, Comisión Nacional de los Mercados y la Competencia (CNMC), Barcelona, Spain

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## ABSTRACT

The launch of ChatGPT in late 2022 has generalized the availability of chatbots with generative artificial intelligence capabilities for end users. From here, the question is whether this technology is being used equally by all user groups or rather whether we may be at the beginning of a new digital divide. The present study addresses this question by using a well-designed, nationwide survey with data on AI chatbot use. Binary and ordered logit regression models are estimated for the actual AI chatbot use. The results show that the older, the less educated and women lag behind in the use of this new technology. In addition, users who make greater use of other digital technologies are also more likely to use generative AI. Although generative AI is a new technology, its pattern of use among individuals seems to follow the path of previous digital divides. The implications of these findings are discussed.

## 1. Introduction

ChatGPT was launched on the November 30, 2022 by OpenAI. It was a significant breakthrough in AI language models as it could conduct natural conversation on almost any topic in the way humans do and became the first AI conversational agent (or chatbot) available to the general public. In the following months other AI chatbots were launched in the wake of ChatGPT. In December 2022, Perplexity (a company founded in August 2022) released Perplexity Ask, their AI answer engine, and the first versions of LLaMa (Meta) and Bard (Google) were released in the first quarter of 2023.

All these AI technologies were distributed to the general population, who could use them in computers and mobile devices and at no cost in some versions. They use advanced large language models primarily based on giant neural networks, specifically a type called transformer models (Vaswani et al., 2017) and are characterized by their ability to understand human text and queries and respond to these interactions by producing responses and content delivered with a “human-alike” tone and speed. The more advanced versions seem to simulate complex reasoning (OpenAI, 2024). These innovations are at the core of what has been defined as generative AI. Given the vast media coverage of these conversational agents, their users are cognizant that they are engaging with an AI, which is different to the situation when an AI is embodied in a platform, device or customer support service.

Since their launch, the number of users of these technologies has grown speedily. According to a poll carried out by IPSOS (2023), six months after the release of ChatGPT, 16 % of Americans had used some type of generative AI chatbot. Pew Research (2024) reports that 18 % of US adults had used ChatGPT by July 2023, a figure that grew to 23 % by February 2024. The statistics about use in Europe are scarce and patchy but are also suggestive of an important growth rate of AI empowered conversational tools. For example, GfK

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<sup>\*</sup> Corresponding author.

E-mail address: [david.suarez@cnmc.es](mailto:david.suarez@cnmc.es) (D. Suárez).

DAM (2023) estimates that in October 2023, more than 4 million Spanish residents used ChatGPT, raising from the 1,4 million it reported in December 2022. Our own research reports that by June 2024, over 20 % of Spanish residents had used AI empowered conversational agents.

Therefore, all these AI chatbots are gaining importance in the personal and professional activities of a growing number of users, yet not much is known about who these users are, a question that is dealt with in our contribution.

On the one hand, identifying the characteristics of these users will enable the detection of the social groups that are at the risk of missing on the different opportunities that AI creates. Certain social groups may be at a disadvantage because of their lack of exposure to this new technology, which promises to advance the personal and work spheres of many individuals. There is a growing concern that AI may create the grounds for a new digital divide and make existing disadvantages and inequalities more pervasive (Gonzales, 2024).

On the other hand, it is also important to understand whether those using generative AI are social groups vulnerable to its risks. Among those is the fact that an AI chatbot may provide biased outcomes, which result from existing ones on underlying training data and from human subjectivity when preparing this data (labelling and annotation) or when developing algorithms (Almakinah et al., 2024). AI has been extensively reported to exhibit systematic partiality against certain population groups, for example on the basis of gender and race (Fang et al., 2024; Hofmann et al., 2024; UNESCO, 2024; Yang et al., 2024), and, although less researched, also on the basis of age (Stypińska, 2021). Bias can also be pronounced against underrepresented population groups, such as individuals with disabilities (Glazko et al., 2024). AI can even spread misinformation or manipulate the user (Al-kfairy et al., 2024). Moreover, it may only expose the user to views that are aligned to her own ideas and therefore censor alternative perspectives (i.e., filter bubbles, see, for example, Nechushtai & Lewis, 2019). There are also concerns about potential misuse of AI chatbots by students and scholars (Howell & Potgieter, 2023). Finally, about the use of AI and breaches of data privacy (Palle & Kathala, 2024). Education and some personal maturity can equip individuals to deal with some of these risks; therefore, it is important to detect who the AI users are, and which AI literacy policies should be considered.

All of this is attracting the attention of public administrations and regulators. Thus, supporting human capital and protecting users are at the core of the EU AI strategy that aims to ensure that AI is trustworthy and human centric, so that AI systems are safe, transparent and accountable. Similarly, California announced public initiatives to encourage safe and responsible AI to protect its citizens but has later stopped them. However, there has been a clear increase in AI regulation in the US. Thus, while in 2016 there was only one AI regulation, in 2023 the number of norms reached 25, with an annual growth of over 50 % in that year (Maslej et al., 2024). Recently, Howell et al. (2025) reviewed the current state of telecommunications and internet regulation, including a juxtaposition of the EU and US approaches.

For all the mentioned reasons, it is important to know who is using this new technology to be able to anticipate whether we are on the verge of a new digital divide. To date, the few studies that have researched this issue either did not have representative samples of the entire population or were merely descriptive. Our study, by using a representative sample and fitting advanced statistical models on it, allows us to identify without bias whether such a digital divide exists and, if so, what factors are behind it.

The rest of the article is structured as follows. Section 2 reviews the relevant literature and discusses in depth the shortcomings of previous research. The research methodology, data sources and statistical models are described in Section 3. The results are presented and discussed in Section 4. Finally, the discussion and main implications of the study are presented in Section 5.

## 2. Literature review

The OECD (2001) defined digital divide as the gap between individuals, households, businesses and geographic areas at different socio-economic levels with regards to their opportunities to access information and communication technologies. Since then, the appearance of new digital technologies has led to multiple “digital divides” beyond the original notion which referred to the adoption of Internet access (Gallardo & Whitacre, 2024; Sethasuravich et al., 2024). For example, nowadays there are references to a divide in digital financial services (Nam & Lee, 2023) and in e-government adoption (Gijón et al., 2025).

Although one might think that the problem of the digital divide is solved simply by waiting for the penetration of the new technology to increase over time until the divide disappears, the evidence does not seem to indicate that this is the case. For example, in a study conducted in Spain analyzing the digital divide present in twelve digital services in the period 2007–2019, the authors found that although it seems that the divide narrows with time for some groups, for others, such as the elderly, the uneducated or those with limited digital skills, it does not (Pérez-Amaral et al., 2021). Similarly, the same authors also conducted a separate study in Spain that revealed the persistence of a gender gap in Internet use (Garín-Muñoz et al., 2022). Therefore, it is very relevant to identify whether at the beginning of the adoption of a new technology, such as generative AI, there are signs of a digital divide which may be perpetuated over time with all the implications that this would entail.

Our paper aims at identifying the characteristics of individuals who use AI-empowered conversational agents, which have been made available to the general population in the last two years. Like this we attempt to the understand whether a new digital divide may arise from the use of this type of AI. This is a relatively novel question within the incipient literature about the acceptance of AI technology. In this area of research, the majority of empirical papers address acceptance with information about individuals’ willingness to use AI (declared intentions) or self-reported attitudes towards AI, which are measured on a qualitative scale.

For example, Albarrán-Lozano et al. (2021) studied the acceptance of AI in Spain by using data obtained in 2018 from a representative survey of 6308 Spanish citizens. These individuals responded to several questions about their general perceptions on innovation, including a few about AI and robots. They assessed how the individual’s general opinions on innovation related to their attitudes towards AI and robots. They found out that males hold more favorable views of AI and robots, which is suggestive of a gender gap.

More recently, Kelly et al. (2023) survey the English-written contributions published until 2022, which deal with the acceptance of devices with an embodied AI technology across several industries. They identify 60 articles but find that most of them consider individuals' willingness to buy/use or try such AI-embodied devices rather than focusing on actual use or purchase behavior. Consequently, they recommend that future studies consider real uptake rather than declared intentions.

All this research predates the advent of ChatGPT and other AI chatbots meant for general use and, therefore, has not analyzed their impact on the acceptance of artificial intelligence. Since then, some papers have already appeared investigating the use of ChatGPT in some particular groups. For example, Jo (2023) studied its use by university students and office workers and Andersen et al. (2025) its use by the scientific community.

Quite differently, our paper deals with the use of AI for general purposes beyond the professional environment or specific groups. To our knowledge, the same question has been addressed only by two studies.

First, Daep and Counts (2024) explores the emerging AI digital divide by analyzing state- and county-level search query data for ChatGPT in the United States during the first six months after its release. Their dataset was retrieved from search interactions with Microsoft's Bing search engine. They found that counties with the highest rates of search were relatively more urbanized and had proportionally more educated, more economically advantaged, and more Asian residents than other counties. However, this work has three important limitations: i) the outcome is only a proxy for actual ChatGPT usage, ii) the data is aggregated at the county level and not at the individual level, which could bias the results (Freedman, 1999), and iii) the contribution excludes generative AI chatbots other than ChatGPT. Our study overcomes these three limitations as: i) our outcome variable is actual usage, ii) our data is at the individual level and iii) our data is not limited to the usage of a single AI chatbot.

The second study examines the users' characteristics that predict individuals' usage of ChatGPT (Kacperski et al., 2025). The authors do this by combining web-tracking and survey data of 1376 Germany citizens during the first 11 months after the launch of ChatGPT. They confirm that those more prone to ChatGPT are the young, the highly educated and those using social media. They also report a negative effect of parenthood and being in full time employment but find that gender and income have no effect on ChatGPT use. These authors use a rich set of ChatGPT use variables sourced from their web tracking exercise (usage, visits and adoption). However, they acknowledge serious limitations to their study. First, the sample size of ChatGPT users was very small (161 individuals) affecting the results regarding number of visits and number of weeks of use. Second, the surveyed individuals self-selected into the study and, thus, the representativity of the sample is questioned. Because of this, the authors warn about the limited generality of their study and call for replication by other scholars. Moreover, the authors also query whether their results would hold generally for AI empowered conversational agents, as several were made available in the market after the release of ChatGPT.

In our paper we overcome these limitations, since our results are based on a large and representative sample of Spanish residents which were selected randomly. The sampled individuals responded to a general-purpose survey, which had one question about how frequent their use of AI empowered conversational agents was during the first half of 2024. Whilst our characterization of AI usage is not as rich as in Kacperski et al.'s, the sample size and characteristics allow for more robust results, which are relevant to the Spanish population and quite possibly like those that one would obtain in other European countries.

Additionally, the notion of digital divide resulting from AI is also related to the concept of AI literacy: i.e. the skills needed to make an efficacious use of AI (Celik, 2023). These skills are necessary for individuals to recognize AI technologies, evaluate which AI to use among those available, critically evaluate the results of AI and be conscious of the ethical considerations surrounding its use (Wang et al., 2022). Several papers assess which user groups are better equipped to use AI in an effective way and will be less exposed to the risks inherent in their use. An interesting contribution (Wang et al., 2024) investigates the digital skills and attitudes towards AI of 1088 Dutch citizens in 2021 and concludes that the elderly and less educated are the most vulnerable to the risks of using AI.

Ultimately usage should have an impact on an individual's AI literacy: Fox and Connolly (2018) express that a smaller willingness to use the technology may ultimately result in lower competence. Indeed, if vulnerable groups are also less prone to engage in AI, this will limit their learning by doing and motivation to acquire relevant skills, which will put them at a disadvantage in the long run. Whilst we do not deal with AI literacy in a direct way, our results can help identify who these groups are.

Finally, another variable that may contribute to a digital divide in the global use of generative AI is language (Kshetri, 2024a). The most common large language models, such as ChatGPT, have been trained using the digital resources available on the world wide web. These contents are mostly in English, so it is not surprising that their performance is best when using this language. Other Western languages such as Spanish or French, and some Eastern languages such as Mandarin Chinese or Japanese, have a smaller volume but still sufficient for large language models to have a performance close to that of English (Kshetri, 2024b). However, this is not the case for other low-resource languages (Choudhury, 2023). As a consequence, AI systems clearly perform worse for other languages than they do for English, even for some with millions of speakers, such as Arabic, Bengali, Hindi, or Urdu (Ahuja et al., 2023). These languages do not benefit from having sufficient digital available content, are penalized by using non-Latin scripts, and in many cases are backed up by governments or corporations with insufficient resources to access the expensive computational architectures needed to train a large language model.

Moreover, perhaps unsurprisingly given the strong tie between language and cultural values, there is also evidence that AI tends to exhibit Western values and worldviews. For example, Tao et al. (2024) conducted a disaggregated evaluation of cultural bias for five widely used large language models and concluded that they reflect cultural values similar to English-speaking and Protestant European countries. At an individual level, this may misrepresent or disconnect users in other contexts from their actual cultural environment and make AI tools less relevant for these groups. At a global level, this may spur a force towards homogenization, potentially reducing diversity.

### 3. Methodology

#### 3.1. Data

Our study draws on a dataset from a survey of Spanish households owned by the Spanish National Markets and Competition Commission (the public body in charge of competition policy enforcement and regulation of utilities in Spain). The fieldwork was undertaken in the second quarter of 2024.

The sample was carefully designed to be representative of the households in Spain. Stratified random sampling was used to select households in each region in Spain according to the size (number of members) of the household, the size (number of inhabitants) of the municipality where the household was located and the socioeconomic status of the household. The resulting sample included persons residing in over one thousand municipalities.

Finally, the present study uses the information provided by 7398 individuals aged 18 or more.

#### 3.2. Variable definitions and summary statistics

Our dependent variable was the use of AI chatbots. Specifically, interviewees were asked: “In the last six months have you used any artificial intelligence chatbot (e.g. ChatGPT, Perplexity, Google Bard/Gemini, etc.)?” and the response options were provided by means of a Likert scale: i) Yes, I have used it regularly, ii) Yes, I have used it occasionally, iii) Yes, I tried it but I stopped using it, and, iv) No, I have not used it. 23.3 % responded affirmatively, this is, either responded options i), ii) or iii). Interestingly, almost one in three of these users had used AI chatbots regularly.

Our explanatory variables included socio-demographic variables and variables related to the use of digital technologies. The first ones comprised age, gender, employment status, educational level and the presence of children at home. The second ones comprised the frequency of use (every day, almost every day, every week, less than every week or never) of GPS, social networks and online music services, having a smart speaker at home, and, finally, owning a mobile with the iOS operating system. [Table 1](#) reports the summary statistics.

**Table 1**  
Summary statistics.

| Variable                                                 | N    | Mean  | Std. Dev. | Min | Max |
|----------------------------------------------------------|------|-------|-----------|-----|-----|
| AI chatbot usage: Yes, I have used it regularly          | 7398 | 0.072 | 0.259     | 0   | 1   |
| AI chatbot usage: Yes, I have used it on occasion        | 7398 | 0.114 | 0.318     | 0   | 1   |
| AI chatbot usage: Yes, I tried it but I stopped using it | 7398 | 0.047 | 0.211     | 0   | 1   |
| AI chatbot usage: No, I have not used it                 | 7398 | 0.767 | 0.423     | 0   | 1   |
| Age                                                      | 7398 | 48.4  | 16.3      | 18  | 91  |
| Gender: Female                                           | 7398 | 0.514 | 0.500     | 0   | 1   |
| Employment: Full time                                    | 7398 | 0.501 | 0.500     | 0   | 1   |
| Employment: Part time                                    | 7398 | 0.078 | 0.268     | 0   | 1   |
| Employment: Retired                                      | 7398 | 0.198 | 0.399     | 0   | 1   |
| Employment: Unemployed                                   | 7398 | 0.075 | 0.264     | 0   | 1   |
| Employment: Student                                      | 7398 | 0.068 | 0.252     | 0   | 1   |
| Employment: Non-employed                                 | 7398 | 0.079 | 0.270     | 0   | 1   |
| Education: Primary                                       | 7395 | 0.110 | 0.313     | 0   | 1   |
| Education: Secondary                                     | 7395 | 0.520 | 0.500     | 0   | 1   |
| Education: University 3-years degree                     | 7395 | 0.153 | 0.360     | 0   | 1   |
| Education: University 5-years degree                     | 7395 | 0.217 | 0.412     | 0   | 1   |
| Children at home: Yes                                    | 7398 | 0.253 | 0.435     | 0   | 1   |
| Use of GPS: Every day                                    | 7341 | 0.125 | 0.330     | 0   | 1   |
| Use of GPS: Almost every day                             | 7341 | 0.155 | 0.362     | 0   | 1   |
| Use of GPS: Every week                                   | 7341 | 0.183 | 0.386     | 0   | 1   |
| Use of GPS: Less than every week                         | 7341 | 0.386 | 0.487     | 0   | 1   |
| Use of GPS: Never                                        | 7341 | 0.151 | 0.358     | 0   | 1   |
| Use of social networks: Every day                        | 7364 | 0.517 | 0.500     | 0   | 1   |
| Use of social networks: Almost every day                 | 7364 | 0.187 | 0.390     | 0   | 1   |
| Use of social networks: Every week                       | 7364 | 0.064 | 0.245     | 0   | 1   |
| Use of social networks: Less than every week             | 7364 | 0.071 | 0.257     | 0   | 1   |
| Use of social networks: Never                            | 7364 | 0.161 | 0.368     | 0   | 1   |
| Use of online music: Every day                           | 7364 | 0.208 | 0.406     | 0   | 1   |
| Use of online music: Almost every day                    | 7364 | 0.184 | 0.388     | 0   | 1   |
| Use of online music: Every week                          | 7364 | 0.168 | 0.374     | 0   | 1   |
| Use of online music: Less than every week                | 7364 | 0.196 | 0.397     | 0   | 1   |
| Use of online music: Never                               | 7364 | 0.244 | 0.429     | 0   | 1   |
| iOS mobile operating system: Yes                         | 7398 | 0.199 | 0.399     | 0   | 1   |
| Smart speaker at home: Yes                               | 7398 | 0.334 | 0.472     | 0   | 1   |

### 3.3. Model specification

Our primary goal was to estimate the drivers of using AI chatbots, irrespective of the intensity of use. Thus, we can uncover which factors impact on the potential digital divide for AI chatbots use. Bearing this in mind, we used a binary logit regression model where the dependent variable was the use of AI chatbots (yes vs. no), where yes was constructed by considering any affirmative answer (options i, ii, or iii described above). The explanatory variables were both the socio-demographic traits and the variables related to digital technologies described in the previous sub-section. Moreover, the models were fitted including the Spanish regions as covariates.

In addition, as the survey responses to the question about the use of AI chatbots were provided by means of an ordinal categorical variable, from regular use to no use at all, we fitted, as well, an ordinal logit regression by using as dependent variable this Likert variable and as explanatory variables, the ones used in the binary model.

To estimate the parameters in both models, we used the maximum likelihood estimation method; then we exponentiated the estimated coefficients to obtain the corresponding odds ratios. On top of that, we used robust standard errors.

Finally, to ensure the reliability and validity of the results of the different models, we used several diagnostic tests. First, we employed the Omnibus tests of model coefficients. Second, we derived several pseudo- $R^2$  coefficients: the Nagelkerke, the McFadden, the Horowitz, and the Cox and Snell coefficients. Third, we estimated variance inflation factors (VIF) values to check for multicollinearity between the regressors. Lastly, to check the binary logit model's performance in classifying individuals into those who use AI chatbots and those who do not, we derived the area under the curve (AUC) corresponding to the receiver operating characteristic (ROC) curve.

## 4. Results

Before commenting on the regression results, it is important to evaluate the fitted models' robustness (Table 2).

The omnibus tests of model coefficients were statistically significant ( $p < 0.001$ ), rejecting the null hypotheses that the models'

**Table 2**  
Diagnostic tests for binary and ordered logit models for AI chatbot use.

|                                                     |                           |               |
|-----------------------------------------------------|---------------------------|---------------|
| <b>Omnibus tests of models' coefficients</b>        |                           |               |
| Binary model: Chi-square = 887.78, p-value <0.001   |                           |               |
| Ordered model: Chi-square = 959.33, p-value <0.001  |                           |               |
| <b>Area under the curve (AUC for the ROC curve)</b> |                           |               |
| Binary model: 0.804                                 |                           |               |
| <b>Pseudo R-Squared</b>                             |                           |               |
|                                                     | Binary model              | Ordered model |
| Nagelkerke                                          | 0.269                     | 0.229         |
| McFadden                                            | 0.194                     | 0.145         |
| Horowitz                                            | 0.188                     | 0.140         |
| Cox and Snell                                       | 0.159                     | 0.161         |
| <b>Test for multicollinearity</b>                   |                           |               |
|                                                     | Binary and ordered models |               |
| Variable                                            | VIF                       |               |
| Age                                                 | 3.07                      |               |
| Gender: Female                                      | 1.16                      |               |
| Employment: Part time                               | 1.13                      |               |
| Employment: Retired                                 | 1.58                      |               |
| Employment: Unemployed                              | 1.19                      |               |
| Employment: Student                                 | 2.20                      |               |
| Employment: Non-employed                            | 1.23                      |               |
| Education: Secondary                                | 3.16                      |               |
| Education: University 3-years degree                | 2.39                      |               |
| Education: University 5-years degree                | 2.73                      |               |
| Children at home: Yes                               | 1.14                      |               |
| Use of GPS: Almost every day                        | 2.35                      |               |
| Use of GPS: Every week                              | 2.67                      |               |
| Use of GPS: Less than every week                    | 4.03                      |               |
| Use of GPS: Never                                   | 3.12                      |               |
| Use of social networks: Almost every day            | 1.22                      |               |
| Use of social networks: Every week                  | 1.11                      |               |
| Use of social networks: Less than every week        | 1.15                      |               |
| Use of social networks: Never                       | 1.44                      |               |
| Use of online music: Almost every day               | 1.87                      |               |
| Use of online music: Every week                     | 1.89                      |               |
| Use of online music: Less than every week           | 2.31                      |               |
| Use of online music: Never                          | 2.83                      |               |
| iOS mobile operating system: Yes                    | 1.09                      |               |
| Smart speaker at home: Yes                          | 1.07                      |               |
| Regional dummies                                    | Min 1.06, Max 1.69        |               |

coefficients were equal to null. To interpret the pseudo- $R^2$  coefficients, we followed the benchmarks provided by Hemmert et al. (2018). The four pseudo- $R^2$  coefficients that we obtained were all within the ranges of a “good or excellent model fit”. In the case of the pseudo- $R^2$  coefficients for the ordered logit model the figures were slightly lower but similar. The maximum value of VIF in the regressors was 4.03, below the usual maximum thresholds (5 or 10) thought to indicate a concerning level of multicollinearity (James et al., 2021). Additionally, the AUC was slightly above 0.8, the cutoff considered as evidence that the model provides excellent discrimination (Hosmer Jr et al., 2013). All in all, these numerous tests provide strong evidence of the accuracy and reliability of our logit regression results.

Table 3 reports the results of both the binary and the ordered logit regression analyses, presenting odds ratios, associated robust standard errors and 95 % confidence intervals for the regressors. Interestingly, the results of the two models are practically the same in terms of both statistical significance and effect sizes.

We found that age had a significant negative effect on the likelihood to use generative AI, thus, the younger you are the more likely you are to use AI chatbots. This result agrees with the findings of Kacperski et al. (2025) who found that age impacts negatively on the usage and adoption of ChatGPT. This result is problematic for two reasons. First, all demographic projections agree that the aging of the world's population is almost a given, so this trend could slow the adoption of generative AI globally, and second, previous research on digital divides shows that the age-related digital divide does not improve over time (Pérez-Amaral et al., 2021). Moreover, the elderly could be the group most likely to benefit from the use of generative AI applied to the world of care and health (Ayers et al., 2023).

In our study, gender had a significant effect on the likelihood to use AI chatbots: the odds of using those were approximately a third lower for women than for men. This result does not align with the results of Kacperski and colleagues (2025) who found no gender impacts on either usage or adoption of ChatGPT. However, our finding agrees well with the result obtained by Albarrán-Lozano et al. (2021) that report a gender gap in positive attitude towards AI. If a digital divide in the use of generative AI by women is confirmed, this could exacerbate the wage gap that disadvantages women as evidence points to an improvement in the productivity of workers using generative AI (Brynjolfsson et al., 2025; Noy & Zhang, 2023).

In relation to activity/employment type our results show that students are nearly as twice as likely to use AI chatbots than full-time employees. In the case of part-time employees, we also find an increase in usage with respect to full-time ones, albeit a smaller one, with a 35 % increase in the odds of use. The fact that students are among the group of early adopters of AI could be good news given the

**Table 3**  
Estimations of binary and ordered logit models for AI chatbot use.

| Variable                                     | Binary logit model |                     |          |          | Ordered logit model |                     |          |          |
|----------------------------------------------|--------------------|---------------------|----------|----------|---------------------|---------------------|----------|----------|
|                                              | Odds ratio         | Robust Stand. error | 95 % LCI | 95 % UCI | Odds ratio          | Robust Stand. error | 95 % LCI | 95 % UCI |
| Age                                          | 0.971***           | 0.004               | 0.964    | 0.979    | 0.972***            | 0.004               | 0.964    | 0.979    |
| Gender: Female                               | 0.675***           | 0.050               | 0.583    | 0.780    | 0.687***            | 0.050               | 0.596    | 0.792    |
| Employment: Full time (ref.)                 | 1                  |                     |          |          | 1                   |                     |          |          |
| Employment: Part time                        | 1.352**            | 0.176               | 1.046    | 1.746    | 1.367**             | 0.175               | 1.064    | 1.756    |
| Employment: Retired                          | 0.972              | 0.151               | 0.717    | 1.319    | 0.964               | 0.149               | 0.712    | 1.305    |
| Employment: Unemployed                       | 1.080              | 0.137               | 0.842    | 1.385    | 1.072               | 0.134               | 0.839    | 1.370    |
| Employment: Student                          | 1.745***           | 0.247               | 1.321    | 2.303    | 1.726***            | 0.234               | 1.324    | 2.250    |
| Employment: Non-employed                     | 1.008              | 0.207               | 0.675    | 1.506    | 1.034               | 0.216               | 0.687    | 1.559    |
| Education: Primary (ref.)                    | 1                  |                     |          |          | 1                   |                     |          |          |
| Education: Secondary                         | 1.009              | 0.129               | 0.785    | 1.295    | 0.930               | 0.117               | 0.727    | 1.190    |
| Education: University 3-years degree         | 1.723***           | 0.261               | 1.281    | 2.319    | 1.531***            | 0.225               | 1.149    | 2.042    |
| Education: University 5-years degree         | 2.958***           | 0.424               | 2.233    | 3.918    | 2.685***            | 0.375               | 2.043    | 3.530    |
| Children at home: Yes                        | 0.845*             | 0.081               | 0.701    | 1.019    | 0.894               | 0.086               | 0.740    | 1.079    |
| Use of GPS: Every day (ref.)                 | 1                  |                     |          |          | 1                   |                     |          |          |
| Use of GPS: Almost every day                 | 0.910              | 0.112               | 0.714    | 1.159    | 0.897               | 0.107               | 0.710    | 1.133    |
| Use of GPS: Every week                       | 0.800*             | 0.097               | 0.631    | 1.014    | 0.782**             | 0.092               | 0.620    | 0.985    |
| Use of GPS: Less than every week             | 0.464***           | 0.054               | 0.369    | 0.582    | 0.456***            | 0.052               | 0.364    | 0.571    |
| Use of GPS: Never                            | 0.328***           | 0.056               | 0.234    | 0.459    | 0.317***            | 0.054               | 0.227    | 0.443    |
| Use of social networks: Every day (ref.)     | 1                  |                     |          |          | 1                   |                     |          |          |
| Use of social networks: Almost every day     | 0.778***           | 0.075               | 0.644    | 0.940    | 0.776***            | 0.073               | 0.645    | 0.934    |
| Use of social networks: Every week           | 0.898              | 0.126               | 0.681    | 1.183    | 0.874               | 0.119               | 0.670    | 1.140    |
| Use of social networks: Less than every week | 0.947              | 0.135               | 0.716    | 1.253    | 0.928               | 0.130               | 0.704    | 1.222    |
| Use of social networks: Never                | 0.559***           | 0.077               | 0.426    | 0.732    | 0.562***            | 0.077               | 0.429    | 0.735    |
| Use of online music: Every day (ref.)        | 1                  |                     |          |          | 1                   |                     |          |          |
| Use of online music: Almost every day        | 1.042              | 0.105               | 0.854    | 1.270    | 1.037               | 0.102               | 0.856    | 1.257    |
| Use of online music: Every week              | 0.736***           | 0.081               | 0.593    | 0.914    | 0.748***            | 0.081               | 0.605    | 0.926    |
| Use of online music: Less than every week    | 0.693***           | 0.077               | 0.557    | 0.862    | 0.702***            | 0.077               | 0.566    | 0.870    |
| Use of online music: Never                   | 0.323***           | 0.043               | 0.248    | 0.420    | 0.328***            | 0.044               | 0.252    | 0.426    |
| iOS mobile operating system: Yes             | 1.265***           | 0.105               | 1.075    | 1.488    | 1.288***            | 0.105               | 1.097    | 1.512    |
| Smart speaker at home: Yes                   | 1.287***           | 0.095               | 1.112    | 1.488    | 1.245***            | 0.091               | 1.079    | 1.436    |
| Regional dummies                             | Yes                |                     |          |          | Yes                 |                     |          |          |
| Observations                                 | 7314               |                     |          |          | 7314                |                     |          |          |

\* Significant at 10 %. \*\* Significant at 5 %. \*\*\* Significant at 1 %. LCI: lower confidence interval. UCI: upper confidence interval.

possibilities that generative AI offers to improve the educational process (Cooper, 2023).

Education was one of the variables with a biggest effect size among the analyzed: individuals with 5-year university studies were almost three times more likely to use AI chatbots than those with only a primary school education. It is interesting to note that this is also the case in the study of Daepf and Counts (2024) where the authors found that education emerged as the strongest predictor of interest in ChatGPT. Moreover, Kacperski et al. (2025) uncover similar results: that higher levels of education are positively related with ChatGPT usage. As we have noted regarding age, this could be problematic since the evidence suggests that the digital divide caused by education does not tend to disappear over time (Pérez-Amaral et al., 2021).

Finally, the presence of children at home slightly decreased the chances to use AI chatbots though the level of statistical significance of this variable was weak.

Let's focus now on the technology variables. For both GPS and online music use, we observe a clear gradient in the results: the more frequent the use of these services, the higher the odds of using chatbots. In the case of social networks, the results were not as clear, although individuals who stated that they do not use social networks were half as likely to use AI chatbots as those who use social networks more intensively. Lastly, for individuals who had a smart speaker at home and also for those who had an iPhone, the likelihood of using AI chatbots increased by about 30 %. Overall, we can interpret these results as showing that having access to and/or knowledge of these modern technologies predisposes one to be a user of the generative AI.

## 5. Discussion

With the irruption of generative AI, AI itself has entered a new phase. Since its beginnings in the 1950s, AI had suffered several winters in which it seemed that its initial goals were unattainable, but all of this changed with generative AI. The impact of this new technology may lead us to the gates of a new industrial revolution with implications in the labor (Brynjolfsson et al., 2025; Noy & Zhang, 2023), educational (Cooper, 2023) and social spheres (Porter & Machery, 2024), surpassing even the impact of the arrival of the Internet at the end of the 20th century. Even health care can be deeply affected by generative AI (Ayers et al., 2023). Faced with this new reality and its profound implications, it is urgent to prevent a new digital divide arising from disparities in individuals' and groups' access and use of generative AI, like the one brought about by the arrival of the Internet.

Our work shows us that the patterns observed at the beginning of the digital divide in Internet access seem to be repeated in the use of generative AI: women, the oldest and the least educated start to lag behind in the use of this new technology.

First, the lack of interest and use of generative AI by these groups may diminish their job prospects and professional opportunities. Interestingly, some first results have suggested that the productivity gains generated by generative AI are greater for novel and/or low skilled workers (Brynjolfsson et al., 2025; Noy & Zhang, 2023). Hence, this would create an opportunity for these individuals, that would not be realized if they were not exposed to this technology. Second, the fact that certain individuals do not engage with AI implies that the training of the AI models will continue to be affected by bias and, ultimately, result in the discrimination of some groups and individuals, perpetuating all sorts of stereotypes. Wei et al. (2025) explain how this partiality in outcomes may permeate decision making in several spheres, including the professional one (affecting job applications and performance evaluations), the financial one (impacts on credit scores and fraud detection) and several others. The effects of bias in the domain of health are also important. For example, Fisher and Rosella (2022) explain that for public health, inequities can happen since AI may affect research questions, the representativeness of samples, the choice of outcome definitions and the optimization decisions made during algorithm development.

As a result and considering that we are at the beginning of this new revolution, the kind of evidence we obtain suggests that both the public authorities and AI companies should be concerned with these emerging disparities and make efforts to promote the use of generative AI in the groups that could be left behind. In this regard, the government of Singapore has announced that all its citizens aged 40 and over will be able to access a credit of up to 4000 dollars to train in these new technologies. On the other hand, the European Union has given its support to a free online course with the aim of training its citizens in basic AI skills. This training led by the company MinnaLearn and the University of Helsinki has already had more than one million students, 40 % of whom were women, a relevant figure but still indicative of the presence of a gender gap. In addition, Estonia a member state of the European Union, which has been a leader in digital citizenship and e-residency, has just launched the AI Leap, 2025 program that will provide Estonian students and teachers with access to AI-based educational tools and teach them the most effective ways to use them (AI Leap, 2025).

These actions point in the right direction, but further initiatives will be required to increase AI literacy and use and make users aware of the benefits, limitations and dangers of generative AI. Among the latter, users need to be aware that generative AI is not free from providing biased or discriminatory responses (Kekez et al., 2025) and that the privacy of information shared with generative AI could be compromised (Ye et al., 2024).

Our work, based on a well-designed country-level survey, has allowed us to identify the factors behind the use of generative AI. However, future research will have to go further by identifying what are the main uses that users are giving to these AI chatbots: assistance in their studies, in their work projects or simply in general or specific knowledge queries, and on the level of knowledge and proficient use of those systems. Future studies should also address whether the factors behind the digital divide in AI use are the same in countries other than Western developed countries. First, the reduced availability of high-speed broadband and access to IT infrastructure in developing countries may deepen the digital divide between developing and developed countries, including now the access to generative AI (Khowaja et al., 2024). Second, many of the languages spoken in these countries are not being used in the training of most common large language models, leading to further disadvantage. Therefore, initiatives are being developed to make these languages useable in generative AIs, for example the AYA project (Cohere For AI, 2024), the Bloom project (BigScience, 2022) or the AfroLM one (Dossou et al., 2022).

## CRedit authorship contribution statement

**David Suárez:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Begoña García-Mariño:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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